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Well Pad Piping Displacement Beyond Limits Predicted During Design

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Abstract

Within a few years from start of operation in one of the gas fields in Oman, it was observed that well pad piping, of multiple wells, experienced movement higher than the expected range. This issue was tackled to understand better the potential root causes and prevent further movement in existing pads and excessive movement in the follow-up wells. In parallel, stress studies were undertaken to confirm that operation remains safe.

It was concluded that flowline thermal expansion and slug forces were the most probable causes for the movement. This suggests that thermal expansion impact and some operating scenarios have been underestimated during design phase. The flowline has most likely expanded considerably pushing the piping away from the support. Thermal expansion could be constrained or tolerated to some extent, slug forces can be limited by controlling the operating mode. The team recommended to monitor the situation and proposed actions to limit the movement and prevent escalation, which included, assessing acceptability of current stresses based on largest movement detected, proposing changes to current support configuration that would accommodate higher displacements, and limit the stresses experienced by the piping on the Well pad.

The novelty is to adequately capture pipe displacement causes during design/modelling stage and improve mitigation measures in the design and operation for future wells.

Keywords: Well Pad, piping displacement, Root Cause Analysis

Introduction

Pipelines and piping systems are critical components in industrial operations, designed to transport fluids safely and efficiently. These systems are subjected to various internal and external factors that can influence their movement, such as thermal expansion, slug forces during transient operating modes, and other operational conditions. To mitigate the risks associated with such movements, piping is typically equipped with pipe supports. These supports are engineered to constrain axial and lateral movements while allowing for expected displacements determined through modeling and analysis. The modeling process accounts for the credible internal and external factors to ensure the integrity and reliability of the piping system.

Despite the careful design and implementation of pipe supports, unexpected pipe displacement has been observed in multiple well pads across different parts of the field. These displacements exceeded the

allowable limits defined during the design phase, potentially posing a risk to the structural integrity of the piping. If such movements extend beyond the design range, they can lead to twisting and damage to the piping, potentially resulting in a loss of containment. The field operation team monitored this movement for a duration of time and noticed that the movement continued to extend further until it exceeded its original expected limit.

Industry practices emphasize the importance of accounting for thermal expansion, transient forces, environmental impact and other operational factors in the design of pipe supports. Advanced modeling techniques and stress analysis are commonly employed to predict and mitigate pipe movements. However, instances of unexpected displacement, such as those observed in this case, highlight potential gaps in the application of these methodologies. The observed pipe displacements at similar supports across multiple well pads suggest a recurring issue that is not fully addressed by existing design practices or operational practices. There was a lack of comprehensive understanding of the root causes driving these excessive movements. Additionally, the long-term implications of such displacements on the safety and reliability of operations were unclear.

The primary objective of this investigation is to identify the root causes of the observed pipe displacements and develop effective mitigation strategies to prevent further movement in existing pads and excessive movement in the future wells. In parallel, stress studies were undertaken to confirm that operation remains safe.

Methodology

A Root Cause Analysis (RCA) was carried out by an integrated team comprising experts from Technical Safety, Piping, and Operation disciplines to determine the potential causes of the pipe movement. The primary objective of the RCA is to prevent the recurrence of such failures by identifying and addressing the root cause, rather than merely addressing the symptoms of the issue. In this case, the analysis aimed to prevent further pipe movement in the existing well pads, which could potentially lead to severe consequences, and to ensure that planned future well pads are better equipped to avoid similar issues.

The RCA process followed a structured methodology, beginning with a clear identification of the problem. The team compiled all relevant data, including operational records, design specifications, and field observations, to support the analysis. Tools such as fishbone diagrams were employed to organize and evaluate the potential causes systematically. These tools allowed the team to group and prioritize contributing factors, ensuring a comprehensive investigation. During the analysis, the integrated team defined potential causes that contributed to the pipe displacement. Each defined cause was systematically investigated separately by assigned action parties. Where applicable, the missing data were planned for collection from the field to support the investigations. For example, field measurements of pipe displacement, support conditions, and environmental factors were gathered to provide additional insights. Causes that could not justify the pipe displacement, based on clear evidence, were eliminated from further consideration. The remaining potential causes were kept for further analysis until they could be conclusively ruled out.

Relevant Data

During the design phase of the project, a piping/pipeline stress analysis report was prepared to evaluate the mechanical integrity and ensure the safety of the piping system. This analysis thoroughly addressed a range of factors, as outlined in the introduction section. The primary objective of the report was to establish an optimal design for the piping system by accounting for all internal and external stresses (forces). This approach aimed to ensure that the piping system would operate safely and reliably under varied operating conditions.

The piping configuration is identical on all the pads except the overall orientation that can change from pad to pad. The piping configuration is shown below in Fig. 1 where the flowline is represented in yellow between the wellhead and the transition to underground flowline.

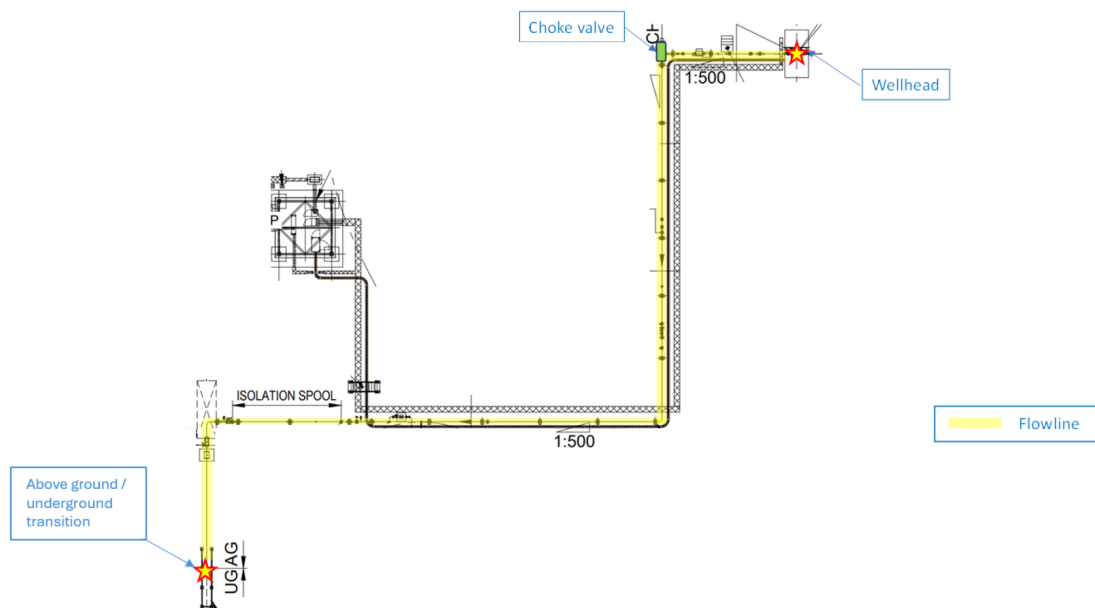


Figure 1—Well pad piping layout from the wellhead to the entry underground (top view)

Between the 2 points from the wellhead to the AG/UG transition, there are 20 supports present. These supports are indicated in Fig. 2 and Fig. 3 below.

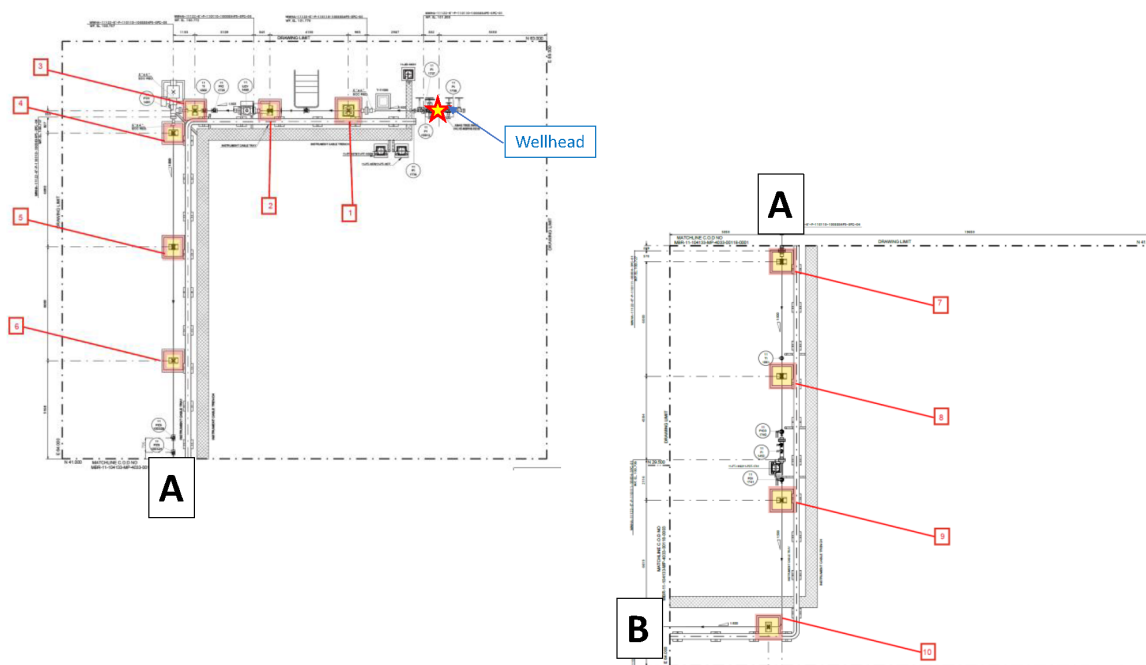


Figure 2—overview of the supports location

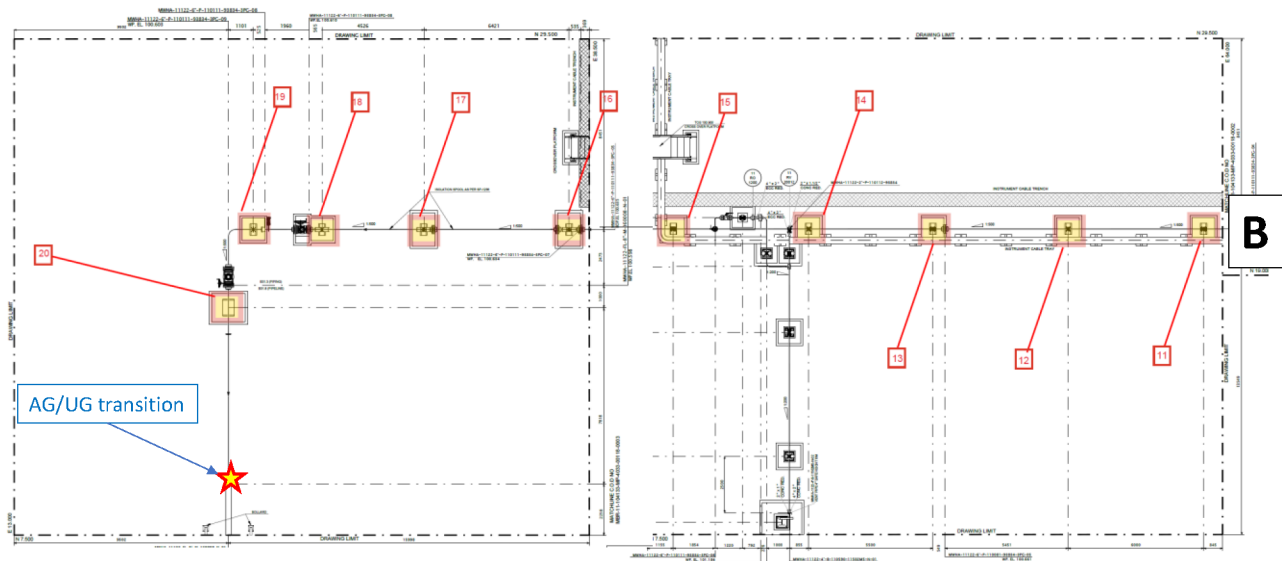


Figure 3—overview of the supports location (cont'd)

The different types of support along the length of the piping are shown in the Fig. 4 below. Some allow for minimal displacement in the axial/lateral direction, other prevent axial/lateral movement. Some supports also prevent lifting of the line.

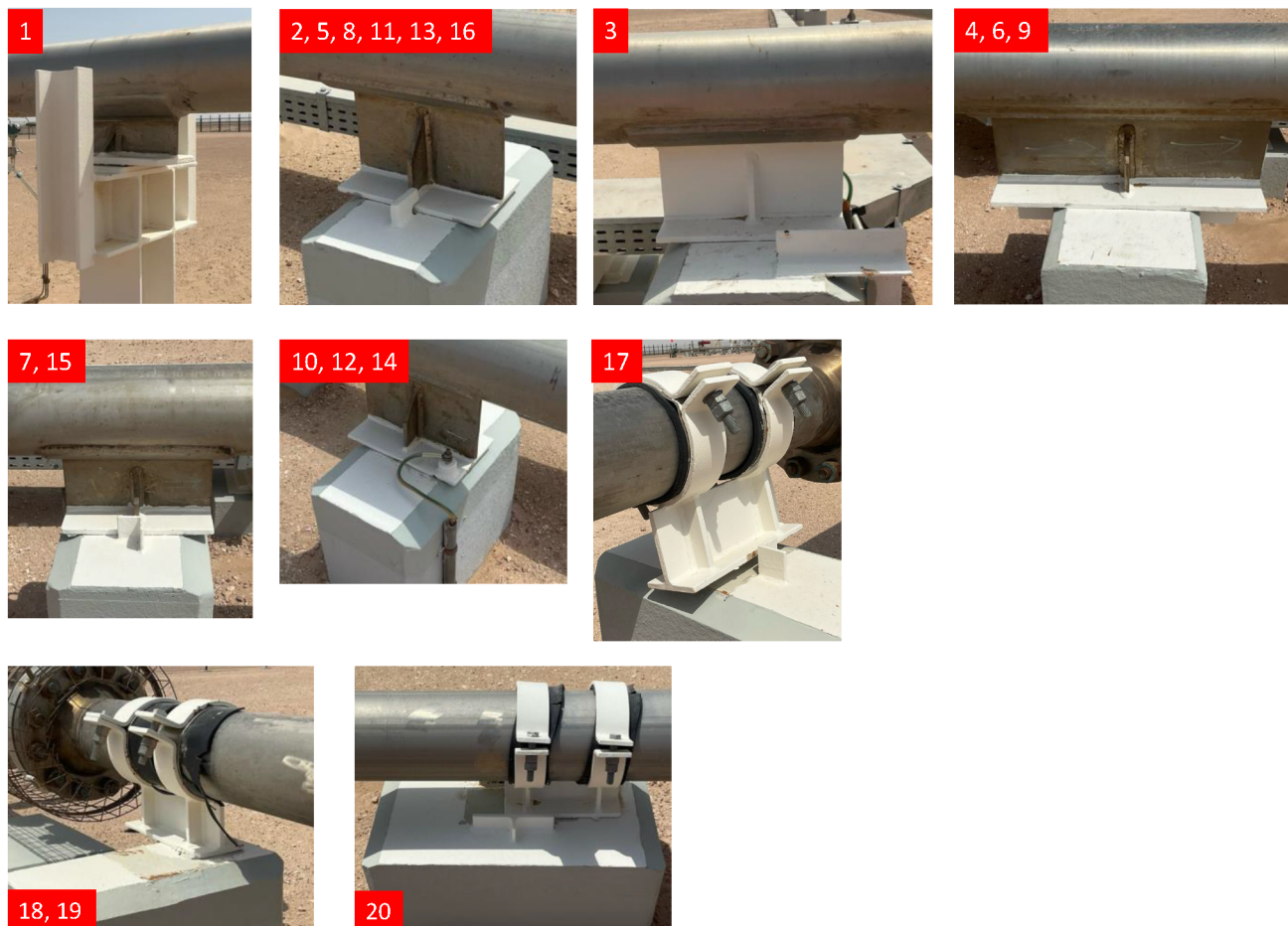


Figure 4—Overview of support types with their location

Based on the stress analysis carried out during design phase, it was determined that there is, on the pipeline going from above ground to underground, an expected axial displacement of 70 mm towards the pad and an expected lift up by 22 mm. It was recommended in the report to bury the line to 2 m for the first ~300 m then reduce the burial depth to 0.8 m all the way to manifold and dive again to 2 m burial depth ~300m before arriving at the manifold. The intent was to prevent extension of the pipeline towards the surface and constrain it under ground.

However, it was reported that the displacement measured at support #19 for the pads have varied between 30 to 200 mm, which exceeds the expected displacement calculated in the stress analysis, regardless of the burial recommendation being applied. In some cases, the supports have twisted, moving away the pipe from the centre of the support concrete panel. The only significant movement detected, in all pads, is in the part downstream of the support point #16.

On the pads, line movements have been detected only from support #17 onwards. This is a pattern which has been seen on a number of pads. Because the piping cannot move sideways at support #17, the piping got twisted as shown below in Fig. 5 and Fig. 6.



Figure 5—Piping twisted on support #17 from one of the well pads (A)



Figure 6—Piping twisted on support #17 from one of the well pads (B)

Overall, the pipe movement, when exaggerated for clarity, can be shown in Fig. 7 and Fig. 8.

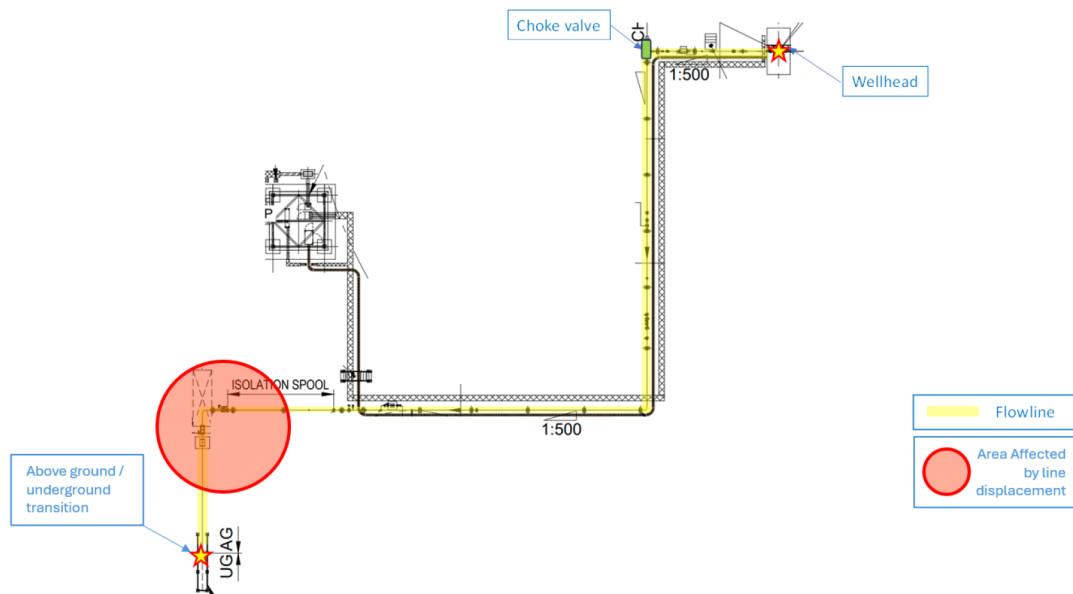


Figure 7—Location on the pad affected by line displacement

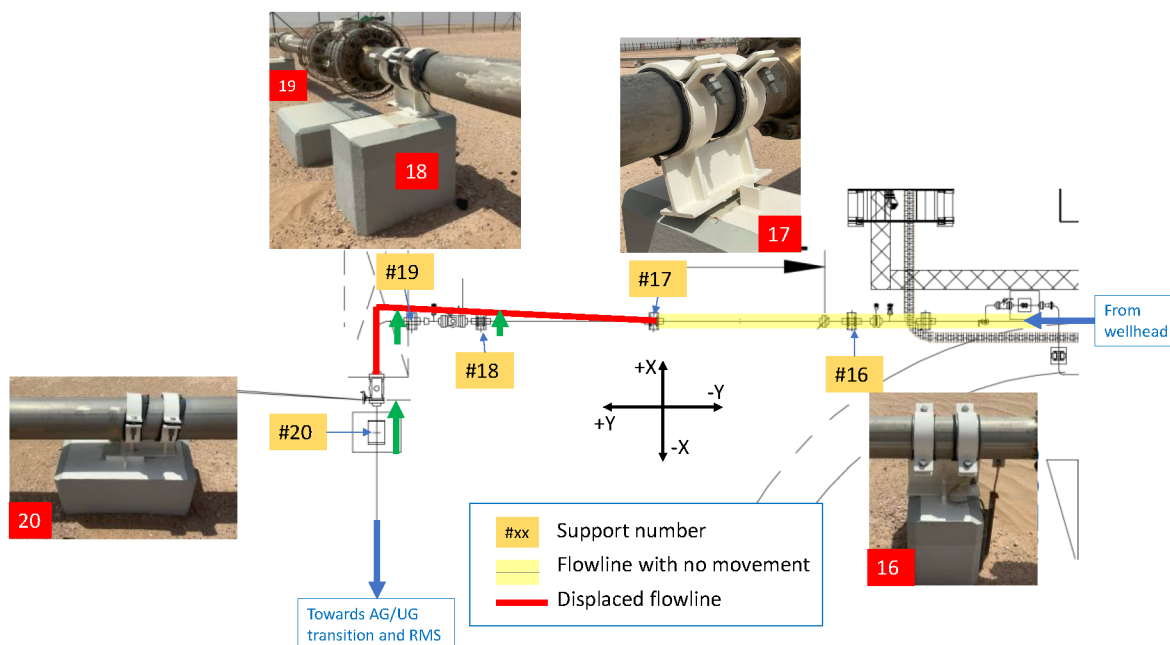


Figure 8—Detail of the location where the line as moved on the well pad

Analysis

The pipe on support #20 moved along the axis of the pipeline and with some movement sideways but not causing the pipe to twist as what is happening on support #17.

The movement of the flowline is caused by a force being exerted on the pipeline. A number of causes for the force have been identified, including both internal and external factors and are listed in Fig. 9 below:

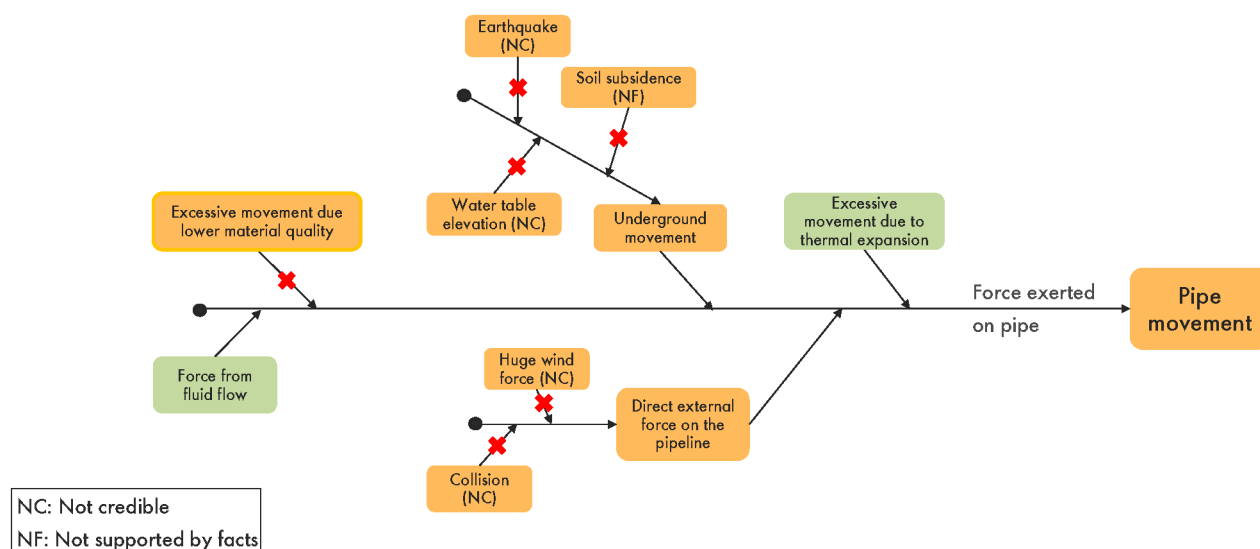


Figure 9—Root causes of pipe movement

Causes that were found to be not credible or not supported by facts have been ruled out and are marked with a red cross. The remaining credible causes i.e. thermal expansion and fluid force are developed further in Fig. 10 and Fig. 11 below.

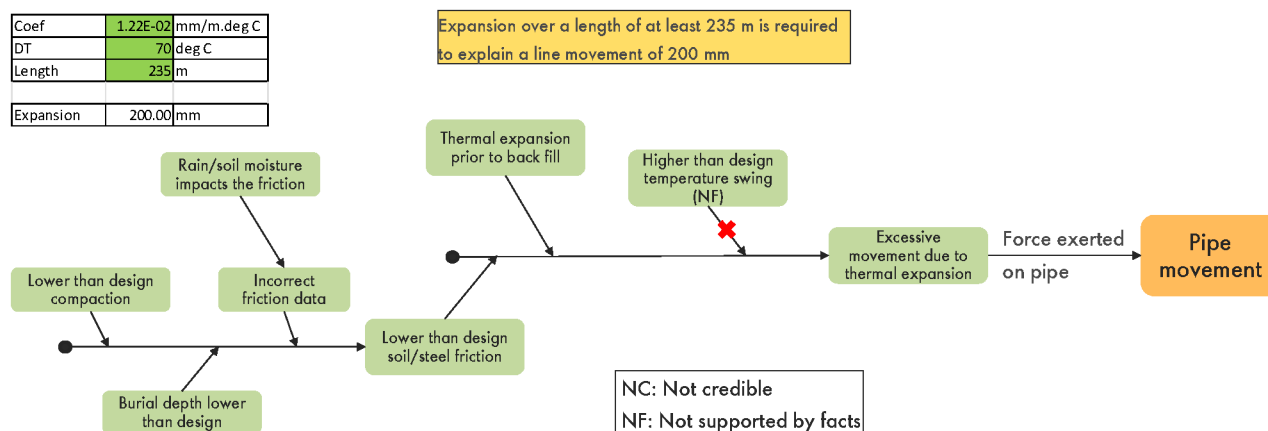


Figure 10—Root cause of thermal expansion

Coef	1.22E-02 mm/m.deg C
DT	70 deg C
Length	235 m
Expansion	200.00 mm

Expansion over a length of at least 235 m is required to explain a line movement of 200 mm

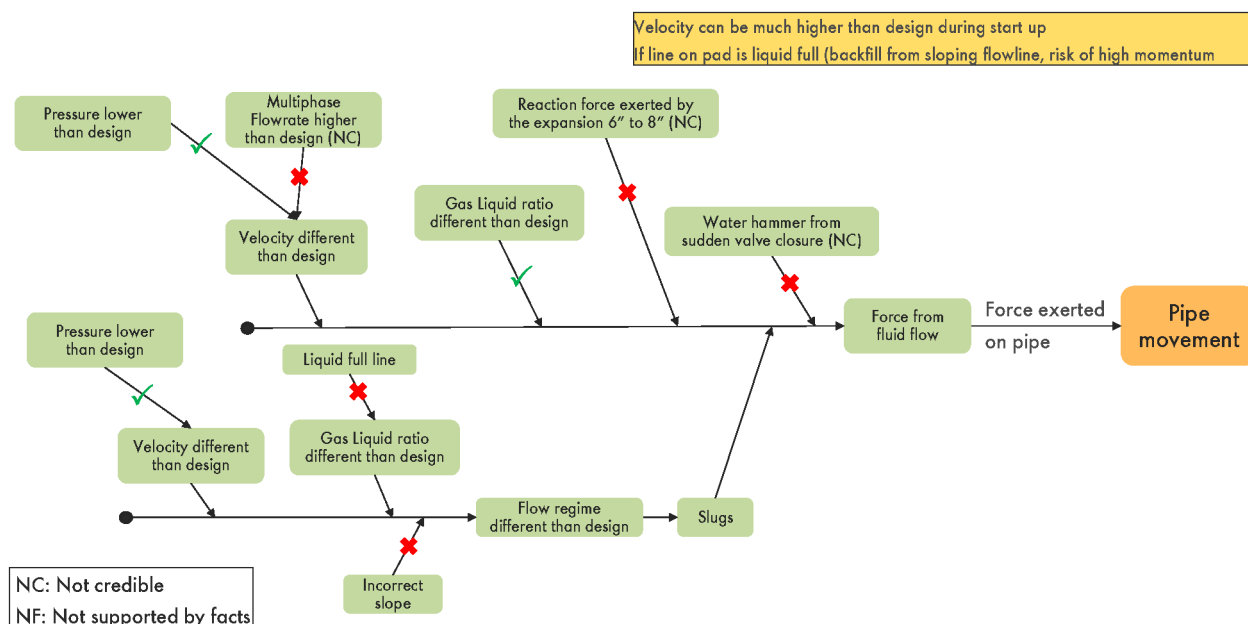


Figure 11—Root cause of fluid force

Thermal expansion

Correlations between parameters

The following graph was produced based on available data of pipe displacement seen on some wells against the well's flowline temperature. This is to examine the relationship between pipe displacement observed in certain wells and the corresponding flowline temperature. The objective was to determine whether internal fluid temperature influences the extent of piping displacement. However, the analysis of the plot indicates that no clear correlation exists between the internal pipe temperature and the observed displacement magnitude.

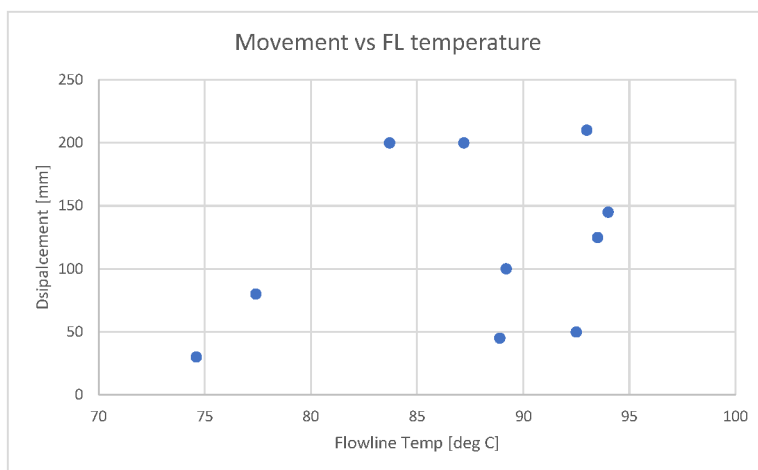
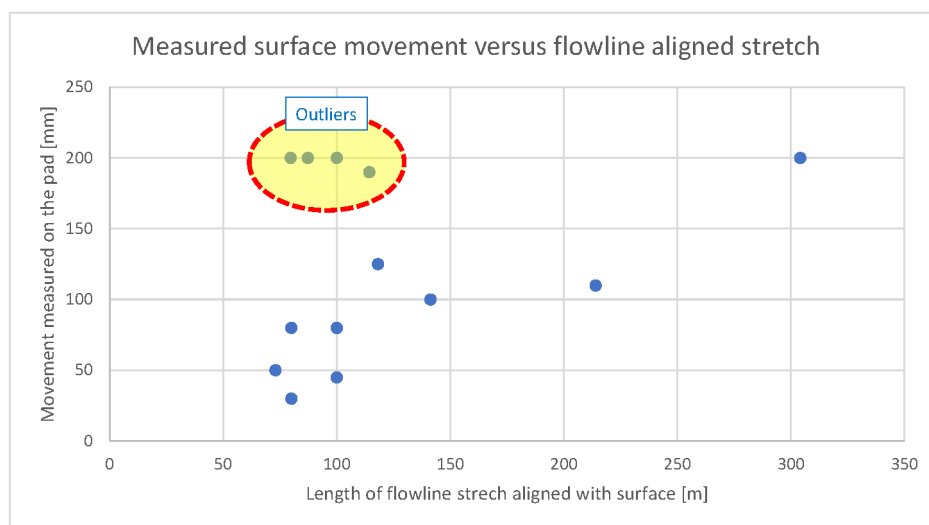


Figure 12—Line movement vs flowline temperature

Another element which is of interest in the context of thermal expansion is the lay out of the pipeline once it has entered the ground. Two examples of flowline routings outside of the well pad (C & D) are presented in Fig. 13 below to highlight the potential difference in impact:



In the case of Well Pad (C), it is apparent that only a small stretch of flowline leaving the pad is aligned with the flowline on the pad, whereas a much longer stretch of pipeline is aligned in the case of Well Pad (D). The expansion of the flowline is more likely to have a significant impact on Well Pad (D) than it would for Well Pad (C). Based on the current measurements of line movement, Fig. 14 is derived.



One notices some logical upward trend, however, 4 well pads, appear to be outliers from this conclusion as they show significant surface piping displacement while the length of straight flowline is in the order of 100 m which would not be enough to explain a 200 mm expansion for the given temperature increase between installation temperature and operating temperature. Hence thermal expansion alone cannot explain the pipe displacement. For that reason, additional investigations have been conducted.

Other givens feeding to the thermal expansion analysis:

- The thermal expansion coefficient of Duplex Stainless Steel, which is the piping and flowline material of construction, is $12.18 \times 10^{-3} \text{ mm/m } ^\circ\text{C}$.

- The temperature variation between the construction temperature and the normal operating temperature is $\sim 70^{\circ}\text{C}$ (96°C operating temperature vs. 26°C installation).
- The length between the fixed points (either UG/AG transition and fixed support #16) and the 90deg elbow is $\sim 15\text{m}$. The thermal expansion will result in a length increase of less than 2 cm.

This means that in order for the thermal expansion to be the cause of the movement, one has to assume that the pipe has been allowed to expand also underground, which means that the soil was not strong enough to hold it in position.

The potential causes for this are indicated in the tree shown in Fig. 10 in the branches "Lower than design soil/steel friction" and "expansion prior to backfill".

Fluid Force

In order to evaluate the forces exerted by the fluid compared to the weight of the system, the evaluation shown in Fig. 15 below has been done, giving a pipe system weight of $\sim 900\text{ kg}$ between the UG/AG transition to support #17.

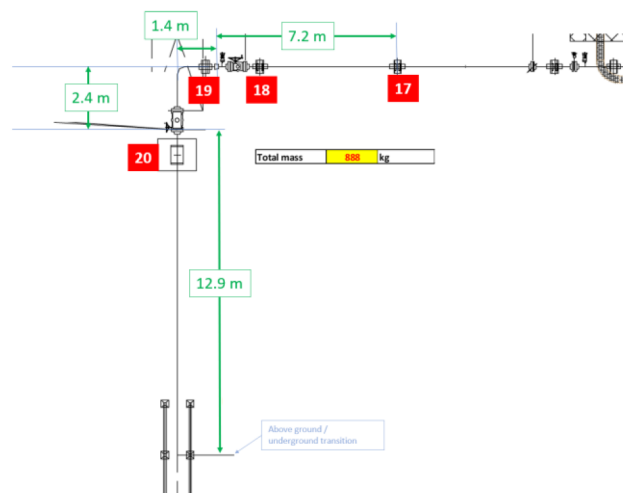


Figure 15—Evaluation of piping system weight

The movement along the axis of the pipe can be either due to the following remaining causes:

- Thermal expansion
- Other axial force on the pipeline due to the slug reaction force

Static force generated by flow and dynamic slug force

The flow and any slug travelling down the flowline will create a reaction force oriented at 45° i.e. having:

- A component pulling the line towards the inside of the pad (green dotted arrow), and
- A component pushing it to the left (blue dotted arrow),

as shown on Fig. 16. The blue dotted component of the slug force will be taken by the fixed support #16 and the red component would pull the pipe inwards, adding its stress to the thermal expansion effect.

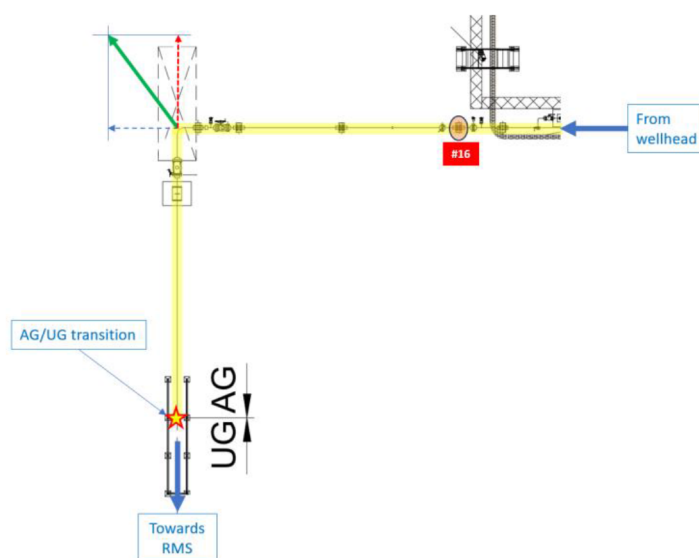


Figure 16—Layout showing the direction of the force created by the flow

The Froude numbers, static and dynamic forces generated by the flow were calculated for credible scenarios to determine the possibility that a flow induced force can move the piping away from the centre of the support.

Scenario 1: Flow regime and fluid force during normal operation

For an 8" piping and wall thickness of 12.7 mm, the calculated maximum reaction force equals 236 N and static force is 78 N. The forces during normal operation are marginal compared to the weight of the piping (ref. Fig. 15). Sensitivities were done at significantly higher CGR but the impact on the force remains marginal and cannot be the cause of significant change in force.

Based on the relationship between gas Froude number and liquid Froude number, the flow regime is determined to be stratified wavy.

Scenario 2: Flow regime and fluid force during initial start up

For an 8" piping and wall thickness of 12.7 mm, the calculated maximum reaction force equals 5,528 N and static force is 433 N. The forces during initial startup are higher, compared to the normal operation case due to higher velocities, but are still modest compared to the weight of the piping.

Based on the relationship between gas Froude number and liquid Froude number, the flow regime is determined to be stratified wavy.

Scenario 4: Fluid force in case of a liquid full line

It was determined that this cause cannot be the root cause of the movement for some of the wells, because of the continuous sloping of the flowline away from the pad, as explained in section.

However, for many wells, the line is not sloping downwards all the way to the manifold. This can be the case, in particular, when the elevation of the manifold is higher than the elevation of the pad. If liquid flows back on the pad, one could imagine, although this is unlikely because the flowline pressure is normally high, that during restart the liquid can be pushed out at high-speed creating some high forces.

Flowline not sloping away from the pad

When the flowline is not sloping away from the pad but has points above the well pad elevation, one cannot exclude that some liquid can flow back on the well pad piping. This was looked at in more detail for one of the well pads. The alignment sheets were used to determine the flowline profile which is reproduced below in Fig. 17, where the different alignment sheets are represented in different colours, while a simplified

profile, drawn in red was used in OLGA dynamic simulation. OLGA was used to determine if, after the shutdown of the well, the liquid present in the flowline can flowback and accumulate in the well pad piping. If that is the case, then the slug forces during restart could be significant.

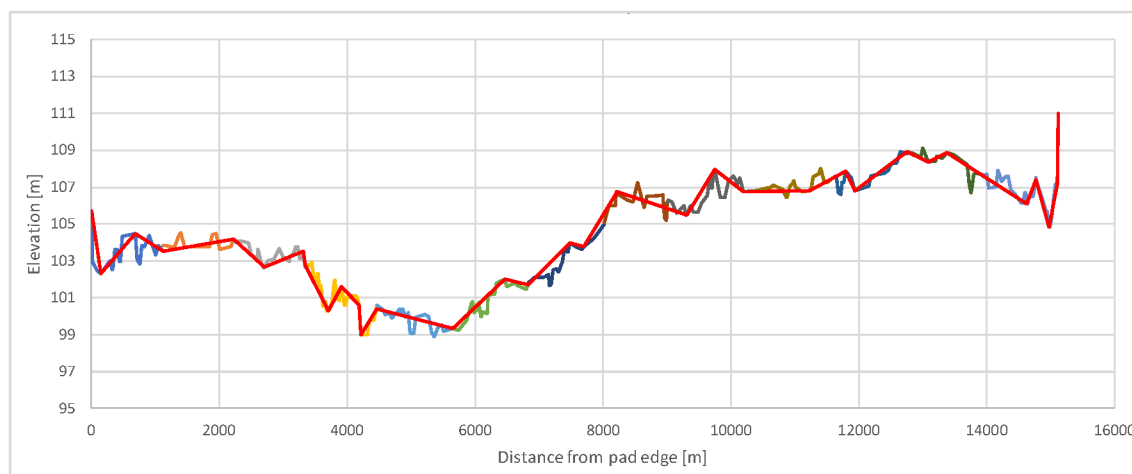


Figure 17—Well pad (A) flowline elevation profile

The liquid holdup across the flowline was modelled in OLGA for the scenario of closing the well, closing the isolation valves located at the edge of the pad and at the inlet to the manifold. After liquid settlement across the flowline, the predicted liquid holdup is as shown in Fig. 18 below:

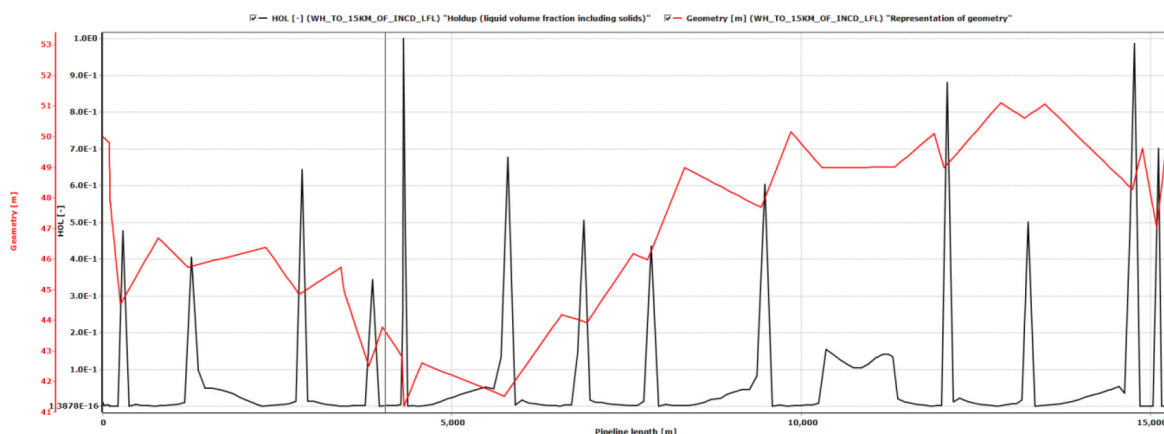


Figure 18—OLGA simulated Liquid Holdup across Well pad (A) Flowline during well shutdown

As per the results, the final liquid holdup profile after liquid settlement across the length of the flowline shows that the flowline does not seem to be liquid full over a large length, making the assumption of filling the well pad piping that could happen in some of the configurations shown in Fig. 21 not very realistic. As per the results seen in Fig. 20, the liquid accumulates at the dips but does not massively flow back towards the pad.

Nevertheless, a full inventory of the elevation of the manifolds compared to the well pads has been done. Based on the inventory, one sees that it is not uncommon that the manifold has a higher elevation than the pad. This still does not prove that liquid can flow back on the well pad but is an indication that it could. There is no clear correlation between the well pad, which are at a lower elevation than the manifold its connected to, and the wells that have been reported to face pipe displacement.

Generally, one can be confronted to the potential following cases in Fig. 19 below, where only in case 1 (the well pad is at a higher elevation than the manifold and flowline has a straight down slope to manifold) is the possibility of backfilling the well pad piping is a very remote scenario.

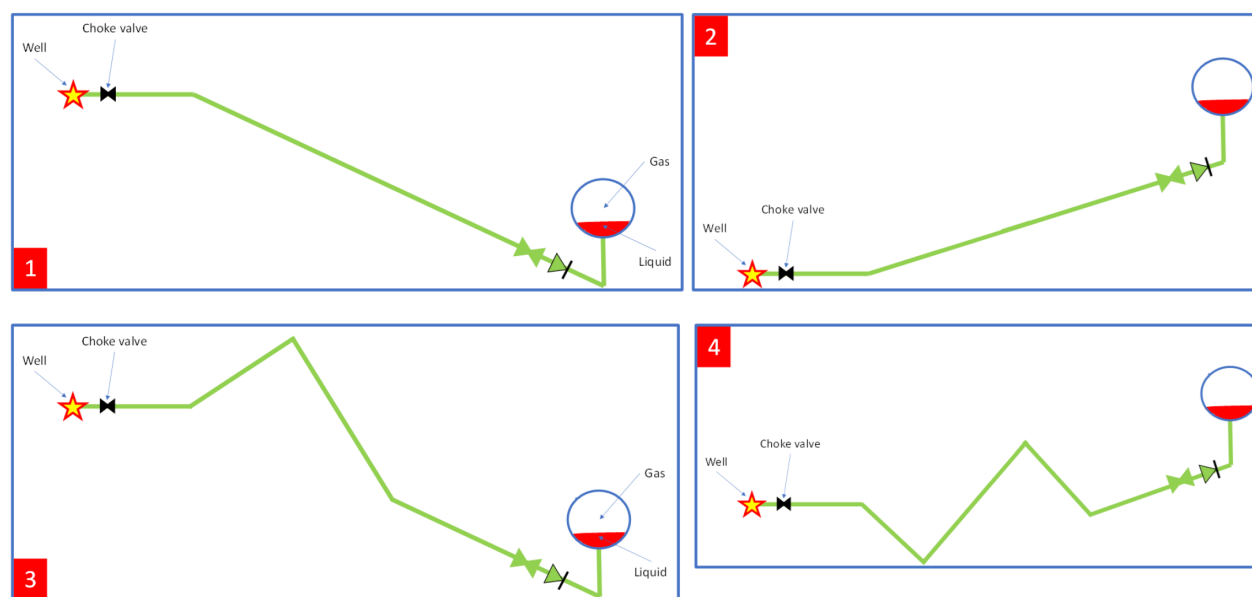


Figure 19—Well pad/manifold potential configurations

In some cases, liquid flowback towards the pad could be possible as shown in Fig. 20, when the well pad is depressurized upstream of the isolation valve before the flowline goes underground. The piping on the pad is depressurized in preparation of maintenance activities at the pad, when a pipe section is required to be worked on. Once the isolation valve is reopened, some liquid accumulated in the flowline can be pushed onto the well pad, making the well pad piping liquid full. If the well start-up rate is too high, the liquid can be pushed out at high velocity creating excessive slug forces on the bends.

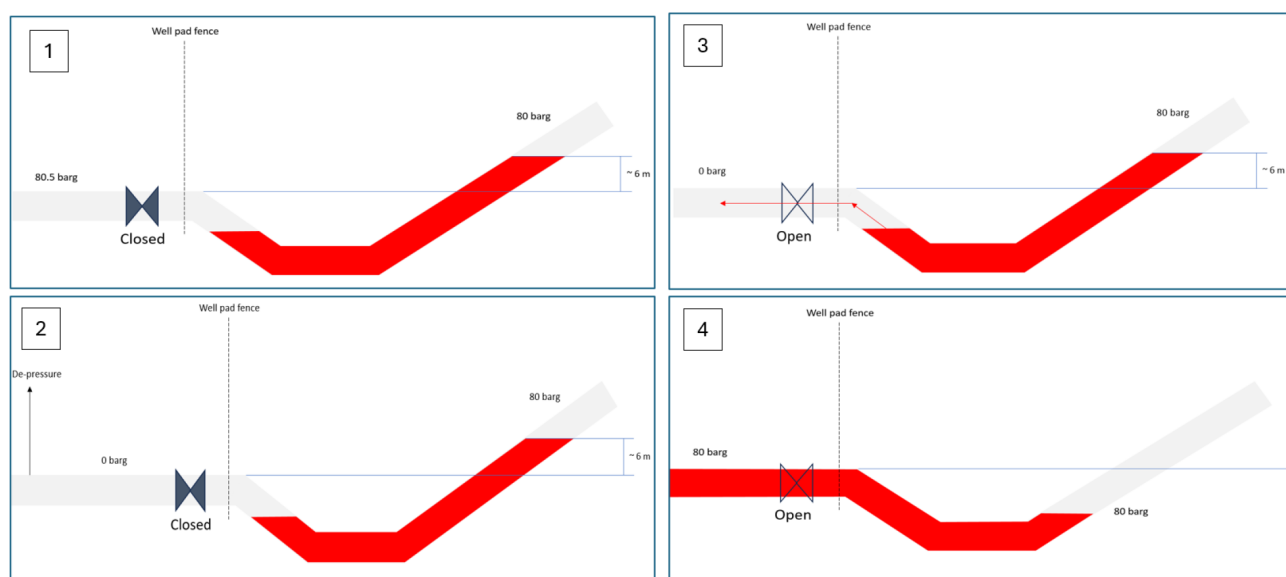


Figure 20—Filling of well pad piping during pad depressurization

As a precaution, the well restart procedure has been updated to include a step during which the choke valve is open only at very small opening to push the liquid back into the flowline at low velocity before the flow is increased further.

Also, it is very unlikely that any slug event causing pipe movement goes undetected during the opening of the valves at the edge of the pad (between the 4 pictures of Fig. 20) because these valves are manually operated and located where the movement is taking place (see Fig. 21).

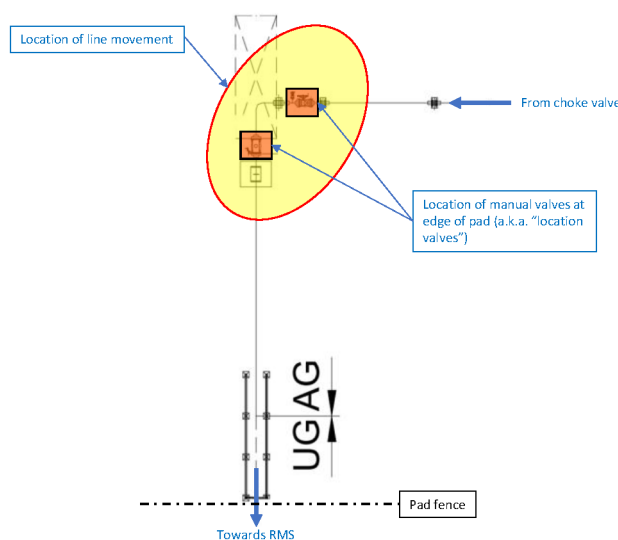


Figure 21—Plot showing location of location valves

Conclusion

This analysis suggests that one of the most probable cause of the observed pipeline movement is thermal expansion, which may not have been adequately captured in the modeling or could have developed prior to the backfilling of the flowline when there were no restrictions to pipeline expansion. Additionally, since thermal expansion alone cannot fully explain the movement, the possibility of liquid presence in the well pad piping has also been investigated.

To address these findings, the following actions were proposed as part of the RCA outcome:

1. Key Factor Identification and Sensitivity Analysis: Identify the critical factors influencing line movement due to thermal expansion, such as lack of compaction, overestimated friction factors, or incorrect line depth. Perform sensitivity analyses to narrow down the most probable cause and propose mitigations or additional measures for Phase 2.
2. Stress Acceptability Assessment: Evaluate the acceptability of current stress levels based on the largest detected movement.
3. Support Configuration Modifications: Propose changes to the current support configuration to accommodate higher displacements and prevent the risk of the well pad piping falling off its support.
4. Revised Stress Analysis: Analyze the stresses experienced by the piping on the Manifold and Central Manifold using a revised stress model that replicates the movements observed on the well pad.
5. Movement Monitoring: Record and monitor pipeline movements across all well pads over time to determine whether the movement has ceased or is ongoing. Adjust monitoring frequency based on criticality and operational experience.
6. Pre-Startup Line Position Recording: Document the pre-startup line position of all wells yet to be started up to better understand the timing of line movement and its potential correlation with well operations.

7. Well Restart Procedure Update: to include in the well restart procedure a step for pushing out any liquid potentially present on the well pad at a low rate before fully opening the choke.

These actions aim to enhance the understanding of pipeline movement mechanisms, mitigate associated risks, and ensure the long-term integrity and reliability of the system.